OUT OF LANE - PROJECT PLAN

Working Weeks	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
Start of the week	2-Dec	9-Dec	16-Dec	23-Dec	30-Dec	6-Jan	13-Jan	20-Jan	27-Jan	3-Feb	10-Feb	17-Feb	24-Feb	3-Mar	10-Mar	17-Mar	24-Mar	31-Mar	7-Apr	14-Apr	21-Apr	28-Apr	5-May	12-May	19-May	26-May	2-Jun	9-Jun	16-Jun	**21-Jun**
Hardware improvements	Mechanical calibration		Mechanical calibration					*Define other necessary sensors, define use-case, integration (IMU, distance), preprocessing, noise cancelling*									*Testing sensors and actuators for proper capture, processing and response*													**BFMC**
	Research GPU addition	Research hardware additions (ultrasonic sensors and headlight)																												
Sensing and input package	Camera handling, ROIs definition							*Define use-cases and test given servers information (localisation on map, cars interaction, gps interaction, interaction with objects)*																						
												Environmental server interaction									*Introduce noise on systems to validate robustness (loss of image, burned image, road search, undefined objects and states)*					*Other functionalities and optimizations*				
Computer vision package (RPi)	Research lane detection		Implement lane detection					*Intersection detection*																						
	Research YOLO for object detection					Implement traffic sign detection		*Implement traffic light detection*			*Trafic lights detection & classification*						*Object detection & classification*													
																										Other functionalities and optimizations				
Behaviour and path planning (RPi)	Robot can be controlled in manual mode			Robot can follow simple instructions																										
Vehicle control package	Lane following and speed control							*Intersection navigation*				*Simple action taking maneuvers (parking, stop for traffic sign, stop for traffic light, stop for pedestrian)*					*Complex action taking maneuvers (swith lane for static and mobile car, road search)*												
																										Other functionalities and optimizations				
Project documentation	Architecture diagram		Architecture revision				Architecture diagram	*Architecture revision*	*Architecture diagram*			*Architecture revision*	*Architecture diagram*								*Project presentation*					*Check Project Presentation*				
	Set up development environment	Embedded system components diagram	Prepare traffic sign dataset					*Define decision making (priorities of actions and state flow)*				*Define robustness and safety measures*																		
	Project plan																									*Other functionalities and optimizations*				
Final result & Demo	Robot can keep a lane, can make a curve							*Robot can navigate in intersection*				*Robot can go on a pre-determined path, stop at stop sign, park at parking sign, slow at crosswalk*					*While detecting and calculation it´s position, the robot can dynamicaly go to specified checkpoint, react to traffic light, interact with other cars ans send enviroment data*				*Robot has full functionality and reacts to any object or situation*							
		Document the 1st report (robot can be controlled remotely)					Document the 2nd report (lane and curve)				*Document the 3nd report (intersections)*									*Document the 4nd report (reactions)*					*Document the 5nd report (complete)*				
																										Other functionalities and optimizations				
Deadlines		15-Dec					19-Jan				16-Feb					17-Mar				20-Apr					21-May					**25/06/2025**
Checkpoint		1st report		Christmas Break			2nd report				3rd report					**Qualifications**				4th report					5th report					Testing - Semi - Finals

Clarification:
Reporting dates have been scheduled for one day before the day specified in the competition schedule, due to the time difference between Romania (UTC +2) and Argentina (UTC -3). This way we ensure that the report is ready when its needed, with a margin of error.